

SAVITRIBAI PHULE PUNE UNIVERSITY



A MINI PROJECT REPORT ON

“Study of Hadoop Ecosystem and its Components”

Submitted by

Name: Adil Hamid Khan

Roll no: 01

CLASS: TE

DIV: A

Under the Guidance of

Ms. Priyanka Jadhav



Sinhgad Institutes

DEPARTMENT OF COMPUTER ENGINEERING
RMD SINHGAD SCHOOL OF ENGINEERING

WARJE, PUNE – 411058

2025 - 26



DEPARTMENT OF COMPUTER ENGINEERING
RMD SINHGAD SCHOOL OF ENGINEERING
WARJE, PUNE – 411058
2025 – 26

CERTIFICATE

This is to certify that the project report entitled

“Study of Hadoop Ecosystem and its Components”

Submitted by

Name: Adil Hamid Khan

PRN No: 72336713C

is a bona fide work carried out by them under the supervision of **Ms. Priyanka Jadhav** and it is submitted towards the partial fulfilment of the requirement of Savitribai Phule Pune University for Third Year.

Ms. Priyanka Jadhav
Guide
Department of Computer Engineering

Dr. Deepali Newaskar
Head,
Department of Computer Engineering

Dr. V. V. Dixit
Principal
RMD Sinhgad School of Engineering Pune – 58

Place: PUNE
Date:

I

Certificate by Guide

This is to certify that **Adil Hamid Khan** has completed the MINI Project work under my guidance and supervision and that, I have verified the work for its originality in documentation, problem statement, implementation and results presented in the Project. Any reproduction of other necessary work is with the prior permission and has given due ownership and included in the references.

Place: PUNE

Date:

Signature of Guide

Ms. Priyanka Jadhav

II

ACKNOWLEDGEMENT

It is my pleasure to acknowledge a sense of gratitude to all those who helped me in making this project.

I thank my Mini Project Guide **Ms. Priyanka Jadhav** for helping me and providing all necessary information regarding this project.

I am also thankful to **Dr. Deepali Newaskar (Head - Department of Computer Engineering)** for providing me the required facilities and helping me while carrying out this project work.

Finally, I wish to thank all my teachers and friends for their constructive comments, suggestions and criticism and all those who directly or indirectly helped me in completing this project.

Adil Hamid Khan

CONTENTS

Certificate	I
Certificate By Guide	II
Acknowledgement	III
1. Problem Statement	1
2. Introduction	2
3. Requirements	3
4. Theory	4
5. Conclusion	10
6. References	11

PROBLEM STATEMENT

The rapid growth of data from various sources such as social media, healthcare systems, and business transactions has led to the need for efficient storage and processing techniques. Traditional systems are not capable of handling such large volumes of structured and unstructured data effectively.

The objective of this study is to understand the Hadoop Ecosystem and its components, including HDFS, YARN, MapReduce, and various tools like Apache Pig, Hive, Spark, and HBase. The system focuses on how these components work together to store, process, and analyze Big Data in a distributed environment.

This study also highlights the importance of Hadoop in real-world applications and demonstrates basic data processing concepts through simple examples.

OBJECTIVES:

- To understand the concept of Big Data and the need for Hadoop.
- To study the architecture and working of Hadoop Ecosystem.
- To understand the role of HDFS in distributed data storage.
- To analyze how YARN manages resources and schedules tasks.
- To study the MapReduce programming model for processing large datasets.
- To explore various Hadoop tools such as Apache Pig, Hive, Spark, HBase, and Mahout.
- To understand real-world applications of Hadoop in different domains.
- To demonstrate basic data processing concepts using a simple example.

INTRODUCTION

In the modern digital era, the amount of data generated from various sources such as social media platforms, healthcare systems, e-commerce, and IoT devices is increasing rapidly. This massive volume of data, commonly known as Big Data, includes structured, semi-structured, and unstructured data. Traditional data processing systems are not capable of efficiently storing and analyzing such large-scale data.

To overcome these limitations, Hadoop was introduced as an open-source framework designed for distributed storage and parallel processing of Big Data. It allows data to be stored across multiple machines and processed simultaneously, which improves performance and scalability.

The Hadoop Ecosystem consists of several components that work together to handle different aspects of Big Data processing. HDFS (Hadoop Distributed File System) is used for storage, while YARN manages system resources. MapReduce is used for processing large datasets in a distributed manner. In addition, tools like Apache Pig and Hive simplify data processing, Apache Spark provides faster in-memory computation, and HBase offers a NoSQL database for real-time data access.

This project focuses on studying these components and understanding how they interact to form a powerful Big Data processing framework. It also highlights the importance of Hadoop in various real-world applications such as healthcare analytics, business intelligence, and data-driven decision making.

REQUIREMENTS

Hardware Requirements:

- a) **Processor:** Intel i3 or higher (i5 recommended for better performance)
- b) **RAM:** Minimum 4 GB (8 GB recommended for handling large data)
- c) **Storage:** At least 500 MB free space (more preferred for Big Data tools)
- d) **System Type:** 64-bit computer or laptop
- e) **Input Devices:** Keyboard and Mouse

Software Requirements:

- a) **Operating System:** Windows 10/11, Linux (Ubuntu preferred), or macOS
- b) **Programming Language:**
 - Java (for Hadoop framework)
 - Python (for demonstration/example programs)
- c) **Big Data Tools (Optional for Practical Understanding):**
 - Hadoop Framework
 - Apache Hive
 - Apache Pig
 - Apache Spark
- d) **Development Environment:**
 - Jupyter Notebook / Google Colab
 - Command Line Interface (CLI)
- e) **Web Browser:** Google Chrome / Microsoft Edge

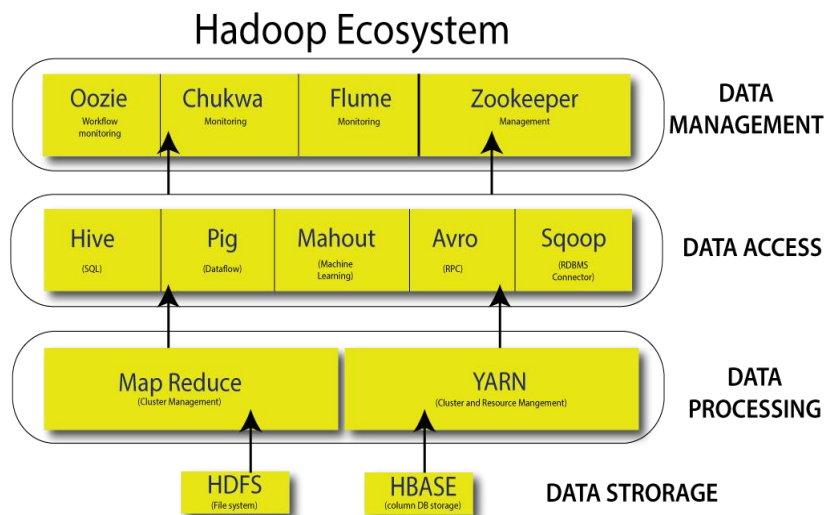
THEORY

1. Hadoop Ecosystem

Hadoop Ecosystem is a collection of open-source tools and frameworks used for storing, processing, and analyzing large volumes of data (Big Data). It provides a distributed computing environment where data is stored across multiple machines and processed in parallel.

The main components of the Hadoop Ecosystem include:

- HDFS for storage
- YARN for resource management
- MapReduce for data processing
- Tools like Hive, Pig, Spark, HBase, and Mahout



2. HDFS (Hadoop Distributed File System)

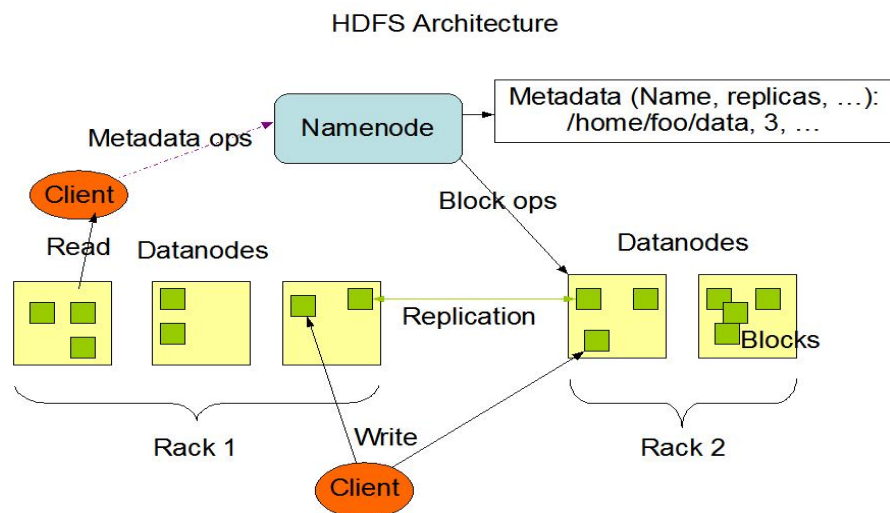
HDFS is the primary storage system of Hadoop. It is designed to store large datasets across multiple machines in a distributed manner.

Components:

- **NameNode:**
Manages metadata and file system structure
- **DataNode:**
Stores actual data blocks

Features:

- Fault tolerance
- High scalability
- Data replication



3. YARN (Yet Another Resource Negotiator)

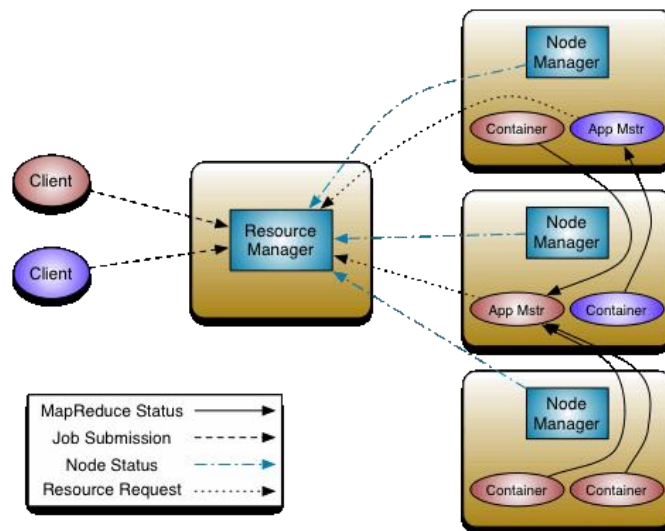
YARN is responsible for managing resources and scheduling tasks in Hadoop.

Components:

- **ResourceManager:**
Allocates resources to applications
- **NodeManager:**
Manages resources on individual nodes

Function:

- Ensures efficient utilization of cluster resources



4. MapReduce

MapReduce is a programming model used for processing large datasets in a distributed environment.

Phases:

1. Map Phase:

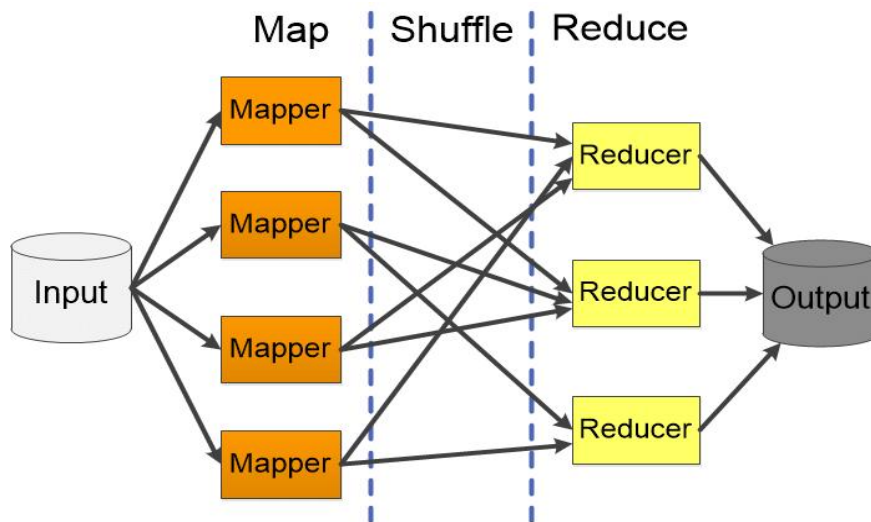
Processes input data and converts it into key-value pairs

2. Reduce Phase:

Aggregates results and produces final output

Advantages:

- Handles large-scale data
- Parallel processing



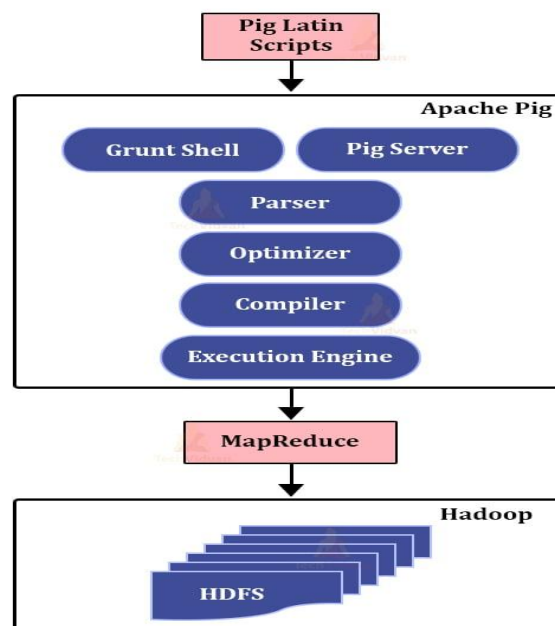
5. Apache Pig

Apache Pig is a high-level data processing tool used in Hadoop.

Features:

- Uses Pig Latin language
- Simplifies complex MapReduce programs
- Suitable for large datasets

Architecture of Apache Pig

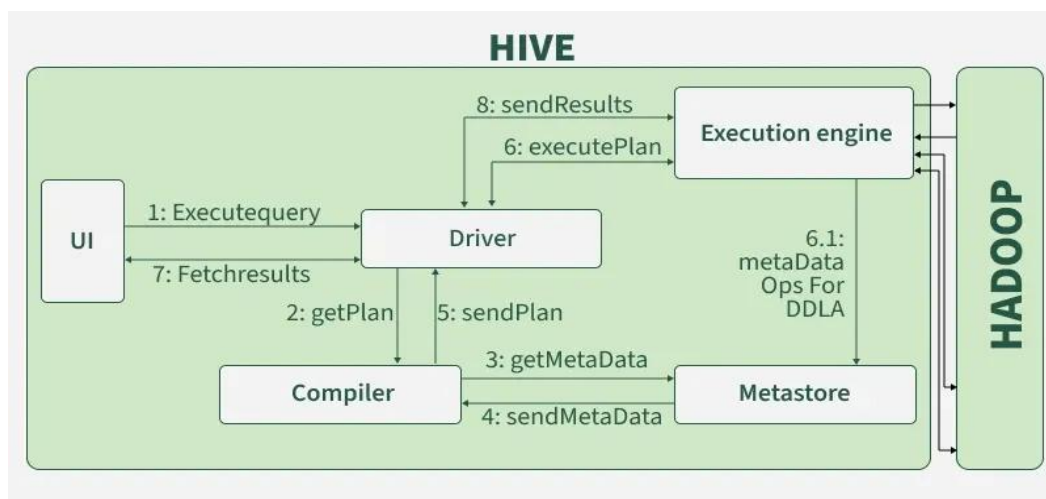


6. Apache Hive

Hive is a data warehouse system built on Hadoop.

Features:

- Uses SQL-like language (HiveQL)
- Easy for querying large datasets
- Used in business analytics



7. Apache Mahout

Mahout is a machine learning library used for analyzing large datasets.

Applications:

- Recommendation systems
- Clustering
- Classification

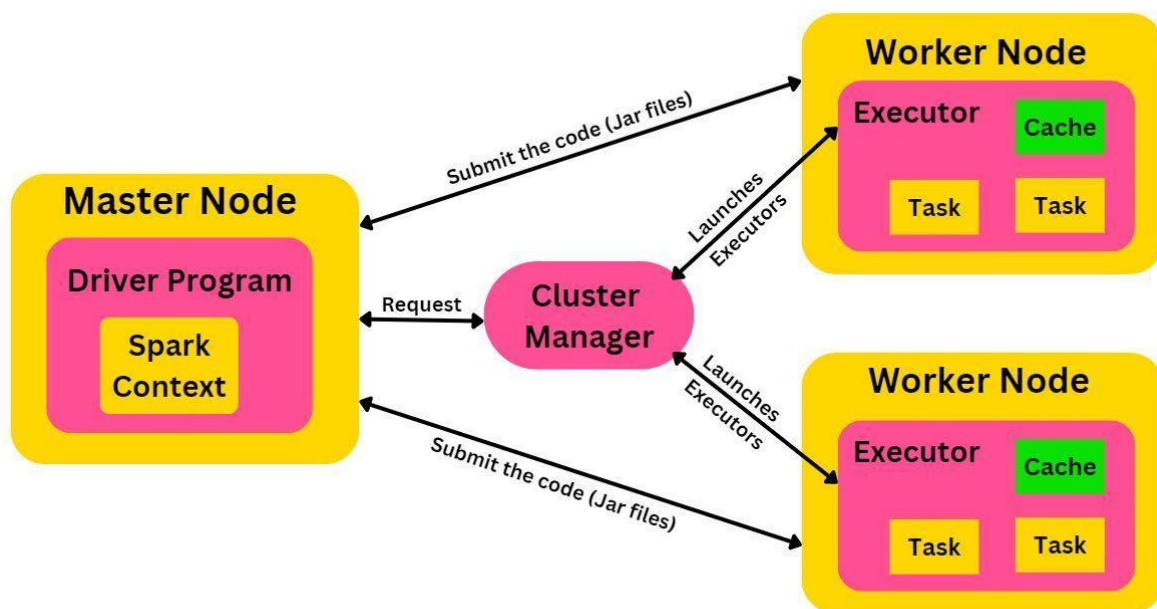


8. Apache Spark

Spark is a fast data processing engine that works on top of Hadoop.

Features:

- In-memory processing
- Faster than MapReduce
- Supports real-time analytics



CONCLUSION

In this study, we explored the Hadoop Ecosystem and its various components used for handling Big Data. The project helped in understanding how large volumes of structured and unstructured data can be efficiently stored and processed using distributed systems.

We studied important components such as HDFS for storage, YARN for resource management, and MapReduce for data processing. Additionally, tools like Hive, Pig, Spark, and HBase were analyzed, which simplify data processing and provide faster and more efficient solutions.

The integration of these components makes Hadoop a powerful and scalable framework for Big Data analytics. It is widely used in real-world applications such as healthcare, business intelligence, and data-driven decision-making.

REFERENCES

- Apache Hadoop Official Website – <https://hadoop.apache.org>
- Apache Spark Documentation – <https://spark.apache.org>
- Apache Hive Documentation – <https://hive.apache.org>
- Apache Pig Documentation – <https://pig.apache.org>
- Apache HBase Documentation – <https://hbase.apache.org>
- NPTEL – Big Data Analytics Course
- GeeksforGeeks – Hadoop and Big Data Tutorials
- Practical Assignment PDF (Group C)